Year 13 Maths - Applications of vectors

Similar questions - Position vectors, perpendicular lines and ratios

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The points A and B have position vectors $\mathbf{i}+3\mathbf{j}$ and $5\mathbf{i}-\mathbf{j}+2\mathbf{k}$ respectively.

(a) Find the length of AB. **[2]**

(b) The point P has position vector $p\mathbf{i}+2\mathbf{j}+2\mathbf{k}$ where p is a constant. Given that angle $APB = 90°$, find the two possible values of p. **[4]**

2 $OADC$ and $BEFG$ are the base and top face of a cuboid with O at the origin, so that $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OC} = \mathbf{c}$ and $\overrightarrow{OB} = \mathbf{b}$, with B above O, E above A, F above D and G above C. The point X lies on AG such that $AX = 2XG$.

(a) Find $\overrightarrow{OX}$ in terms of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$. **[3]**

(b) The point M is the midpoint of AG. Show that the lines OF and AG intersect, and give the position vector of the point of intersection. **[4]**

Question 1

(a) $\overrightarrow{AB} = 4\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$, length $\sqrt{16+16+4} = 6$

(b) $\overrightarrow{PA} = (1-p)\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ and $\overrightarrow{PB} = (5-p)\mathbf{i} - 3\mathbf{j} + 0\mathbf{k}$. Perpendicular: $(1-p)(5-p) - 3 = 0$, i.e. $p^2 - 6p + 2 = 0$, so $p = 3 \pm \sqrt{7}$

Question 2

(a) $\overrightarrow{OG} = \mathbf{b} + \mathbf{c}$. $\overrightarrow{AG} = \mathbf{b} + \mathbf{c} - \mathbf{a}$. $\overrightarrow{OX} = \mathbf{a} + \frac{2}{3}(\mathbf{b} + \mathbf{c} - \mathbf{a}) = \frac{1}{3}\mathbf{a} + \frac{2}{3}\mathbf{b} + \frac{2}{3}\mathbf{c}$

(b) $\overrightarrow{OF} = \mathbf{a} + \mathbf{b} + \mathbf{c}$. A point on OF is $\lambda(\mathbf{a} + \mathbf{b} + \mathbf{c})$; a point on AG is $\mathbf{a} + \mu(\mathbf{b} + \mathbf{c} - \mathbf{a}) = (1-\mu)\mathbf{a} + \mu\mathbf{b} + \mu\mathbf{c}$. Equating coefficients: $\lambda = 1 - \mu = \mu$, so $\mu = \lambda = \frac{1}{2}$, consistent for all three vectors, so the lines intersect at the point with position vector $\frac{1}{2}(\mathbf{a} + \mathbf{b} + \mathbf{c})$ (the midpoint M of AG)